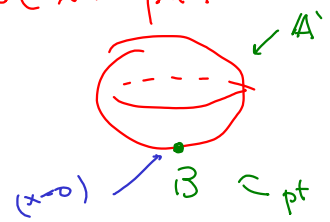
Exercise: (in notes, Friday)

① Work out flag variety $\mathcal{B} \simeq G/B$ (choice of Borel)

$\mathcal{B} = \{\text{Borels of } \mathfrak{g}\}$

Choice of Borel $\rightsquigarrow$ point

SL_2: $\mathcal{B} = \mathbb{P}^1$



B orbits in G/B

$\rightsquigarrow$ Schubert stratification (indexed by W)

Choice of Borel $\Rightarrow$ stratification

Action of $G \curvearrowright G/B$ $\rightsquigarrow$ $\mathfrak{g} \longrightarrow T_x = \mathrm{Lie}(x)$ $\rightsquigarrow$ $U\mathfrak{g} \longrightarrow D(G/B)$

② In category $\mathcal{O}$: $G = SL_2$, 5 indecomposable objects.

Refer to notes for list

Exercise: Work out the corresp. D-modules.

Remark: $\mathcal{O}$: • f.g. • $\mathfrak{z}\mathfrak{g}$, n locally finite • $\mathfrak{h}$ semisimple.

$\mathcal{O}'$: " " • $\mathfrak{z}\mathfrak{g}$ semisimple

$(\mathfrak{z}\mathfrak{g})^*$ Blocks

• generalized eigenvectors

• strict

$\boxed{\text{Guess } P(-2)}$

$\mathcal{O} \longleftarrow D(N^- \backslash G/B)$ fixes choice of B^-, $T = B^- \cap B$

$\downarrow T$

$\mathcal{O}' \longleftarrow D(N \backslash G/B)$

$D(G/B) \simeq \mathcal{O}_{T^*G/B}$ as sheaves.

Compute $\Gamma(G/B, D_{G/B})$

Jeffry: $U\mathfrak{g} \xrightarrow{\alpha} D(G/B) \rightsquigarrow \mathcal{O}(\mathfrak{g}^*) \longrightarrow \mathcal{O}(T^*G/B)$ gr

$\frac{G \times \mathfrak{b}^\perp}{G}$ where $g \cdot (x, b) = (xg, g^{-1}bg)$

$T^*G/B = G \times^B \mathfrak{b}^\perp$

For 2: Springer res.

$\mathfrak{g}^* \leftarrow (g, x) \rightarrow gxg^{-1}$

induces a v.b. on G/B fibers $V_{gB} \simeq gBg^{-1}$

① surjective

② identify the kernel

$(T^*G/B)_{gB} \simeq (\mathfrak{g}/\mathfrak{b})^* \simeq \mathfrak{b}^\perp \underset{\text{Killing}}{\simeq} \mathfrak{n}$

① factors

② $\simeq$ on $\mathcal{O}$.

$T^*G/B = \tilde{\mathcal{N}}$

nilpotent cone

$\mathcal{N} \longrightarrow \mathfrak{g}^* = \mathrm{Spec}\ \mathrm{Sym}\,\mathfrak{g}$

$\{0\} \longrightarrow \mathfrak{g}^*/\!/G = \mathrm{Spec}\,(\mathrm{Sym}\,\mathfrak{g})^G$

Rmk: $\ker(\alpha) = U\mathfrak{g} \otimes_{\mathfrak{z}\mathfrak{g}} k$

Next: $\mathrm{gr}(\mathfrak{z}\mathfrak{g}) = \mathcal{O}(\mathfrak{g}^*)^G$.

Ex: $G = SL_2$

$G/B = \mathbb{P}^1$

$$\begin{tikzcd} \tilde{\mathcal{N}} \arrow[d] & \\ \mathcal{N} \arrow[r, "\Gamma"'] \arrow[d] & \mathfrak{g}^* \arrow[d] \\ \{0\} \arrow[r] & \mathfrak{g}^*/\!/G \end{tikzcd}$$

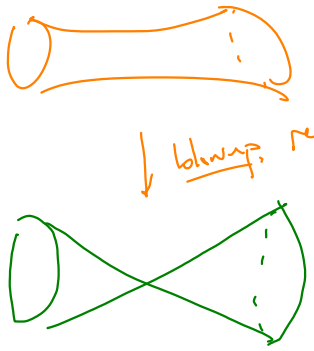
$\mathfrak{g}^*/\!/G$ = parametrize closed G-orbits

$\lambda \to \mathbb{A}^1/\mathbb{Z}/2$ $\quad$ $\begin{pmatrix} \lambda & \\ & -\lambda \end{pmatrix}$

$\lambda^2 \to \mathbb{A}^1$

$x^2 + yz = 0$

$\begin{pmatrix} x & y \\ z & -x \end{pmatrix} \mapsto \boxed{0 = -x^2 - yz}$



blowup, res. of sing.

$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ fixes $\langle e_1 \rangle \subseteq \langle e_1, e_2 \rangle$

$N \cdot e_2 = e_1$

2 orbits

$N = 0$ fixes $\mathbb{P}^1$

Lie theory: • Harish Chandra iso:

$$Z(U\mathfrak{g}) \xrightarrow{\simeq} \mathrm{Sym}(\mathfrak{h})^W \leftarrow \text{shifted}$$

$\rho = \frac{1}{2}\sum_{\alpha \in \Delta^+} \alpha$

$w \cdot (v) = w(v+\rho) - \rho$

"center" is at $-\rho$, not 0

Ex: $\mathfrak{sl}_2$: $\Omega = \frac{1}{8}H^2 + \frac{1}{4}EF + \frac{1}{4}FE$

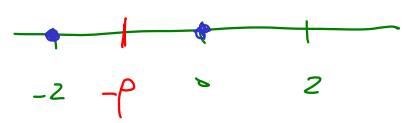
If V is cyclic, Ω acts by scalar (e.g. V highest weight). ~~what scalar?~~

On highest weights, E acts by 0. Push E to right.

$\Omega = \frac{1}{8}H^2 + \frac{1}{4}(FE + H) + \frac{1}{4}FE$ acts by $\frac{1}{8}H^2 + \frac{1}{4}H = \frac{1}{8}(H^2 + 2H)$

$\in \mathrm{Sym}(\mathfrak{h})^W$

Suppose that Ω act by 0. What are the possible highest weights? $H^2 + 2H = 0$ $\quad$ $\boxed{H = 0, -2}$



Thm: (Chevalley rest. thm) $k[\mathfrak{g}^*] \to k[\mathfrak{g}^*]$ usual W-action

Define an equivalence $k[\mathfrak{g}^*]^W \simeq k[\mathfrak{g}^*]^G$

Ex: $\mathfrak{g}^* \to \mathfrak{h}/W$ "eigenvalues" coeff. characteristic polynomial (i.e. S_n-inv functions on eigenvalues

"matrix"

$\mathfrak{h}$ "eigenvalues"

Claim: $\mathcal{Z}\mathfrak{g} \xrightarrow{\simeq} (\mathrm{Sym}\,\mathfrak{h})^W$ $\rightsquigarrow$ $(\mathrm{Sym}\,\mathfrak{g})^G \xrightarrow{\simeq} (\mathrm{Sym}\,\mathfrak{h})^W$

filtered

filtered b/c shifted action

usual action

Lemma: $f: X \to Y$ proper birational map, Y normal.

then $f_*\mathcal{O}_X \simeq \mathcal{O}_Y$

Proof: · local on Y, assume Y affine

· f proper $\Rightarrow$ $f_*\mathcal{O}_X$ f.g. $\mathcal{O}(Y)$-module
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